Do ESG investors care about carbon emissions? Evidence from securitized auto loans*

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Abstract

Securitized auto loans present a clean empirical setting to measure preferences for green assets in price and quantity data. I document three puzzling facts: (i) Brown securities which finance high-emission loans have a 6.5% lower cost of capital. Mutual funds marketed as ESG (ii) hold positions across the full distribution of CO_2 emissions and (iii) invest more in higher-emission deals compared to non-ESG funds. I attribute these findings in part to the use of ESG ratings of issuers instead of direct measures of emissions.

JEL classification: G12, G18, G20, G41, Q56

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I have nothing to disclose.

1 Introduction

Over the last decade, assets managed by funds using environmental, social, and governance (ESG) criteria have increased tenfold. The large inflow of capital into ESG funds over the past decade suggest an appetite from investors for green assets. High demand for green assets could lower the cost of capital for green firms. However, a lower cost of capital is evidence of *non-pecuniary preferences*[1] only if *green* assets have the same risky payoffs as *brown* assets.

Estimating whether investors have non-pecuniary preferences for green assets is challenging: it is difficult to hold expected cash flows constant while varying ESG characteristics. For example, CO_2 emissions may influence expected cash flows across states of the world because of a potential carbon tax. The correlation between expected cash flows and ESG characteristics can introduce unobserved heterogeneity across securities. Open questions about data reliability of ESG rating providers, which strategy ESG investors follow, and whether an activity is ESG or greenwashing further complicate causal inference.

This paper addresses these challenges and sheds light on whether green assets have a lower cost of capital. I compare the cost of capital for senior tranches of green and brown automobile asset-backed securities. Auto ABS are highly standardized debt instruments which securitize pools of auto loans. Only a few parameters distinguish auto ABS deals from each other besides their collateral pool. However, the CO_2 emissions of collateral pools, an objective measure of *greenness*, vary substantially. Moreover, senior tranches of auto ABS are considered safe assets, similar to US treasuries (Gorton, 2017). I exploit the safe-asset nature of senior tranches together with variables derived from loan-level data to hold risk and expected cash flows across securities constant. I then test if CO_2 emissions influences the cost of capital of auto ABS and the decision of ESG investors to invest in them.

My framework allows me to rule out confounders that affect both expected cash flow and ESG-criteria in an effort to identify causal effects. Two features of the securitization market allow me to do this. First, the security design of auto ABS reduces the number of risk factors, with prepayment being the main risk for AAA-rated tranches. Second, prepayment is determined by borrower and loan characteristics rather than the collateral itself. This distinction,

[1] E.g., as in Heinkel, Kraus, and Zechner (2001), Oehmke and Opp (2020), Pástor, Stambaugh, and Taylor (2021), Berk and van Binsbergen (2021), Pedersen, Fitzgibbons, and Pomorski (2021), Zerbib (2022), Goldstein, Kopytov, Shen, and Xiang (2022).

combined with detailed loan-level data, enables me to causally disentangle cash flow from ESG considerations. My identification strategy allows me to answer the following questions: once an investor can be sure that the loan they extend will be paid back at the agreed time, does the environmental impact of the collateral influence their choice to invest and is this choice reflected in prices?

To this end, I collect loan-level data on the collateral pools of all publicly traded auto ABS. The data covers 281 auto ABS deals by 22 issuers from 2017 to 2022, backed by 19 million loans. The data contains information on the borrower, contractual terms, and collateral of each loan. By merging data about the collateral at the make, model, and year level to data on CO_2 emissions from the Environmental Protection Agency, I am able to estimate the lifetime emissions of each vehicle in a collateral pool. The loan-level data allows me to control for the predictors and ex-post performance of collateral pools. Controlling for both predictors and ex-post performance at issuance together with high-dimensional fixed effects removes as much unobserved heterogeneity as possible and allows me to rule out alternative explanations.

I first document large cross-sectional dispersion of CO_2 emissions across auto ABS deals. Auto ABS issued by Honda finance an average of 45t of CO_2 per car, while deals issued by Ford finance an average of 95t per car. I show that the amount of emissions and thus the greenness of an auto ABS deal can be measured by the share of trucks in its collateral pool: a 1pp increase in the share of trucks increases the amount of financed CO_2 by 1.02t per car. This "sufficient statistic" is available to investors in the prospectus and should enable them to distinguish green from brown securities.

Using the cross-sectional dispersion in CO_2 emissions, I show that the effects on the cost of capital depend on whether it is possible to account for heterogeneity across securities. In a *naive* specification that only accounts for market conditions at issuance, auto ABS with emissions below the median have a statistically significant lower cost of capital of around 25%. This difference shrinks towards zero and eventually flips sign when I account for security design choices and exposure to prepayment risk. This finding highlights the importance of being able to account for risk factors and security design, when analyzing the existence and drivers of a premium for green assets.

Instead, in regressions that control for prepayment risk I document that *higher*-emission

collateral pools have a *lower* cost of capital[2]. Auto ABS with emissions above the median have a 6.5% lower cost of capital. Alternatively, moving from the 20th percentile of emissions per US dollar to the 80th percentile (a 1.85 standard deviation distance equivalent to comparing Toyota to Ford) reduces the cost of capital by 6bps. The magnitude of this difference is economically meaningful: it is a 14% lower cost of capital compared to the mean issuance spread of 42bps. The result that brown auto ABS have lower cost of capital is robust across alternative specifications, measures of greenness, and samples.

A potential threat to identification is that CO_2 emissions could be correlated with loan performance. If high emission pools were safer, it would provide a rationale for investors to assign a lower cost of capital to browner securities. I show that emissions do not predict performance at the collateral pool-level or loan-level. This suggests that the lower cost of capital for brown auto ABS is driven by factors unrelated to loan performance, such as investors' non-pecuniary preferences.

I confirm the finding that auto ABS investors have non-pecuniary preferences for certain issuers by showing that higher ESG of *issuers* also cause a lower cost of capital. A 1% higher ESG rating of the ABS issuer is associated with a 0.16% to 0.37% lower cost of capital. Further, ESG ratings of issuers are positively correlated with CO_2 emissions at the collateral pool-level which attenuates the influence of emissions on issuance spreads by up to 50%. This suggests that the negative relationship between spreads and emissions is not driven by a taste for higher CO_2 emissions *per se* but by preferences for issuers with high ESG ratings which produce high emissions collateral.

I next turn to portfolio holdings data of mutual funds to investigate if the CO_2 emissions of the collateral influence ESG investors' decision to invest in auto ABS. In seminal work, Hong and Kacperczyk (2009) argue that norm-constrained investors, like ESG mutual funds, are more likely to invest in responsible securities compared to unconstrained investors. I test this argument in the portfolio data of mutual funds. Mutual funds are key investors in the auto ABS market; up to 85% of a senior tranche is held on their balance sheets.

I leverage that mutual funds hold positions in multiple auto ABS to identify if ESG funds invest more capital in greener auto ABS compared to non-ESG funds. By using a stringent

[2]This stands in contrasts to other results in the literature (e.g, Goss and Roberts, 2011, Chava, 2014, Zerbib, 2019, Larcker and Watts, 2020, Tang and Zhang, 2020, Flammer, 2021, Baker, Bergstresser, Serafeim, and Wurgler, 2022, Fatica, Panzica, and Rancan, 2021, Huynh and Xia, 2021, Seltzer, Starks, and Zhu, 2022, Pástor, Stambaugh, and Taylor, 2022, Aswani and Rajgopal, 2022).

set of fixed effects I measure ESG-funds' preferences for greenness from variation in CO_2 emissions of multiple auto ABS deals held by ESG funds relative to non-ESG funds during the same reporting period. The empirical specifications include collateral pool and fund fixed effects and thus absorb characteristics and preferences of each mutual fund and the specific features of each auto ABS deal. By including these fixed effects, the model controls for as much of the unobserved heterogeneity across collateral pools and funds as possible. However, the relative difference in preferences for greener assets by ESG versus non-ESG funds remains identified.

Two puzzling facts emerge from the analysis of ESG mutual funds holding: ESG funds (i) invest across the full distribution of CO_2 emissions and (ii) have larger portfolio shares in higher-emission deals compared to non-ESG funds. ESG funds allocate approximately 20% less capital to auto ABS deals with emissions below the median than non-ESG funds. Both findings are hard to reconcile with common ESG strategies which usually prescribe outright exclusion or best-in-class investment of brown securities[3].

These puzzling facts are partially explained by the fact that higher emissions pools have higher ESG ratings on average. As expected, ESG funds allocate more capital to auto ABS issued by firms with higher ESG ratings when compared to non-ESG funds. However, ESG funds' reliance on the ESG ratings of auto ABS issuers tilts their portfolio towards higher emissions pools compared to non-ESG funds. When controlling for the ESG ratings of issuers, the impact of CO_2 emissions on the portfolio allocation of ESG funds relative to non-ESG funds becomes attenuated and imprecisely estimated but remains positive.

I innovate in two ways over the literature. First, focusing on senior tranches of auto ABS allows me to isolate the effect of greenness on prices and quantities while holding traditional risk factors like default and prepayment risk constant across securities. Holding risk factors which influence expected cash flows constant is important as prices react to risk caused by greenness[4]. For senior tranches of auto ABS, borrower and loan characteristics determine cash

[3]There is an active debate about which strategy ESG investors should follow: *exit* (divestment or exclusion) versus *voice* (shareholder activism). Broccardo, Hart, and Zingales (2022) analyze the relative effectiveness those strategies. Edmans, Levit, and Schneemeier (2022) analyze whether exclusion or best-in-class investment is more effective.

[4]For example, Seltzer et al. (2022) show that corporate bonds with poor environmental profiles have higher yield spreads, particularly when these corporations' facilities are located in states with strict regulatory enforcement. Regulatory risk caused by poor environmental performance is distinct from ESG preferences affecting prices.

flow risk while a single collateral characteristic, CO_2 emissions, determines their greenness. Separating greenness from cash flow risk allows causal inference about the effect of non-pecuniary preferences on prices and quantities, previously obscured by confounding factors like regulatory risk. Second, I use *continuous* measures of greenness rather than a discrete definition. Previous studies of the green premium in debt markets have focused on municipal and corporate bonds (e.g., Baker et al., 2022, Larcker and Watts, 2020). These bonds are either green if they fund "environmentally friendly'' projects or not green. However, there can be large qualitative and quantitative differences between green projects. Economists do not know yet how investors react to these differences. I address this problem by focusing on a single, continuous dimension of environmental relevance[5].

The paper is organized as follows. The rest of the introduction discusses the related literature. Section 2 provides a brief overview over the ABS market for auto loans and establishes stylized facts about the emissions content of auto ABS. Section 3 describes the data. Section 4 discusses the identification strategy. Section 5 reports estimates of the cost of capital of green and brown auto ABS and discusses results. Section 6 studies ESG preferences in the portfolio holdings of mutual funds. Section 7 concludes.

Related Literature The rise of ESG investing has sparked an active literature[6]. Theoretical contributions highlight that if the share of ESG investors is large, green firms have a lower cost of capital. Heinkel et al. (2001) models an equilibrium where the investment behavior of ESG investors raises the cost of capital of polluting firms and lowers the cost of capital for green firms. Pástor et al. (2021) propose a model that studies the impact of changes in sustainability preferences on asset prices; analyzing equilibrium implications of sustainable investing, welfare, and social impact. Oehmke and Opp (2020) characterize conditions under which socially responsible investors impact firm behavior, in a setting in which firm production generates social costs and is subject to financing constraints. Berk and van Binsbergen (2021) study the theoretical effect of divestment in a single period mean-variance environment; calibrating their model to current data shows that the impact of stock exclusions on a firm's cost of capital is small.

The result of this paper, that browner securities have a lower cost of capital, stands in con-

[5]Note that the auto ABS I study are inherently brown since they finance CO_2 emissions. However, the cross-sectional differences in their emissions, render some auto ABS deals greener than others.

[6]Both Gillan, Koch, and Starks (2021) and Hong and Shore (2023) provide recent literature reviews.

trast to prior empirical results. Pástor et al. (2022) document a near-arbitrage opportunity in which German government green "twin" bonds relative to their non-green twins trade at 5bps lower yields. Baker et al. (2022) estimate a green premium of about 6bps in a sample of over 2,000 U.S. municipal and corporate green bonds, Zerbib (2019) estimates a green premium of 2bps in a sample of over 1,000 supranational, sub-sovereign and agency, municipal, corporate, financial, and covered green bonds. In contrast, Larcker and Watts (2020) find no premium on municipal green bonds; pointing out shortcomings in the estimation methods used in previous studies. My results and the ostensibly conflicting results show that securities that are especially marketed as green trade at a premium while securities that markedly differ in their true greenness but are not marketed as such do not.

I also contribute to the literature on securitization. Klee and Shin (2020) study asymmetric information and signaling in the auto ABS market. Benetton, Mayordomo, and Paravisini (2021) analyze European auto ABS and find that the vertical integration of manufacturing and credit provision allows captive auto lenders to increase cash collected from car sales through credit fire sales.

2 Securitized Auto Loans and their Emissions

This section gives a brief introduction of the market for securitized auto loans, explains key concepts, and establishes several stylized facts. I show that similar to a motorist's choice to buy a high-emissions vehicle or a low-emissions vehicle, an investor has the choice to invest in an auto ABS that finances high-emissions vehicles or relatively low-emissions vehicles. I further show that a simple summary statistic, the share of trucks, is enough to differentiate green from brown deals.

2.1 The ABCs of Auto ABS

Auto loans are one of the most commonly used forms of consumer credit, as automakers find it advantageous to offer loans to increase sales. Auto ABS were among the first consumer ABS to come to the market in the 1980s, when securitization of consumer loans grew popular. At the end of 2021, auto ABS accounted for around 18% of total outstanding auto loans in the US,

with a value of around $220 billion[7]. There are on average 10 new public auto ABS with a total notional value of approximately $13 billion per quarter over my sample period from 2017 to 2022.

Compared to corporate and municipal bond markets, the security design of the auto ABS market is highly standardized. Only a few parameters distinguish auto ABS deals from each other besides their collateral pool. These are the weighted average life of a tranche, credit spread over a yield curve, assumed prepayment speed, and an option to call the bond prematurely. The high levels of standardization in the auto ABS market and the safety of these securities make auto ABS highly liquid as the low bid-ask-spreads of $0.04 and immediate market impact estimates of $0.01 show. For comparison, this is a similar level of liquidity and a lower level of market risk than agency mortgage backed securities that trade in the $200 billion daily volume to-be-announced market (He and Mizrach, 2017).

All 281 deals in my sample are structured as monthly amortizing with higher seniority tranches receiving repayments first. Appendix Figure B1 shows examples of auto ABS deal structures. For a description of the corporate bond underwriting process by investment banks, which is similar to the ABS underwriting process, see Siani (2022). The auto ABS market can be split into prime and sub-prime deals based on the creditworthiness of their underlying loans; sub-prime deals command higher issuance spreads.

Prepayment is the main risk factor for investors in senior tranches of auto ABS since time and risk tranching, high levels of over-collateralization, and other credit enhancements (e.g., reserve accounts or excess spreads) mitigate default risk. Prepayment risk can materialize in several ways, primarily as the early repayment of loans by consumers but also if a borrower defaults on a loan and the car is repossessed. This risk is measured and priced using a prepayment model[8]. Auto ABS deals further have embedded clean-up call options, allowing the issuer to prematurely call the outstanding notes if the remaining pool balance drops below a certain percentage (most commonly 5% or 10%). Clean-up call options, however, are not relevant for most senior tranches since those are paid off before the pool-balance reaches the cutoff point.

[7] see SIFMA, U.S. ABS issuance and outstanding.

[8] Prepayments on auto ABS are measured relative to a prepayment standard or model. The model used in all 281 deals in my sample is the Absolute Prepayment Model (APS). APS assumes a constant rate of prepayment each month relative to the original number of receivables. APS further assumes that all of the receivables are the same size, amortize at the same rate, and that each receivable in each month of its life will either be paid as scheduled or be prepaid in full. For example, in a pool of receivables originally containing 10,000 receivables, a 1% APS rate shows that 100 receivables are expected to be prepaid each month.

While all of the auto ABS deals I analyze are registered securities and publicly traded, there is a considerable amount of auto ABS which are privately placed under Rule 144A. Private placements do not disclose public loan-level information and are excluded from my sample.

2.2 Stylized Facts about CO2 Emissions in auto ABS

My empirical analysis studies the effects of CO_2 emissions of vehicles which serve as collateral for asset-backed securities on issuance prices and quantities purchased by investors. Auto ABSs need to disclose granular loan-level data which allows me to calculate the financed CO_2 emissions for each ABS. The amount of expected emissions that the auto ABS collateral pool b is financing is given by the sum over each vehicle i as:

$$\mathbb{E}\left[\text{Financed } CO_2 \text{ Emissions}\right]_b = \sum_{i \in b} \underbrace{CO_2 \text{ Emissions}_i \times \mathbb{E}\left[\text{Survival-Weighted Miles}\right]_i}_{\text{Expected Emissions}} \times \underbrace{LTV_{0,i} \times \text{Outstanding Balance Share}_i}_{\text{Financing Adjustment}}. \quad (1)$$

The first term of the right-hand side of eq. (1) is the CO_2 emissions of vehicle i measured in tons of CO_2 per mile driven. The second term is the expected survival-weighted vehicle miles traveled over the lifetime of the vehicle. The product of these terms is the total expected lifetime emissions of a new car[9]. The loan-to-value (LTV) part of the financing adjustment of eq. (1) reflects that not all of the expected CO_2 emissions are financed through a loan since many consumers make down-payments at the time of purchase. The financing adjustment further takes into account that loans in bond-pools are warehoused with the original lender for a some time and only securitized after a certain balance has been paid off.

By matching the vehicle collateral to emissions data from the EPA, I am able to calculate the amount of financed emissions for each deal. The CO_2 emissions of ABS deals vary substantially across and within issuers. Figure 1 shows boxplots of the average financed tCO_2 emissions per car across all collateral pools for auto ABS by Honda, Toyota, Santander Bank, General Motors (GM), and Ford. The figure illustrates a striking fact: similar to a motorist's choice to buy a high-emissions vehicle (e.g., Ford F-150) or a low-emissions vehicle (e.g., Toyota Prius),

[9] I adjust the survival-weighted vehicle miles traveled of used vehicles to reflect the remaining expected lifetime of the vehicle.

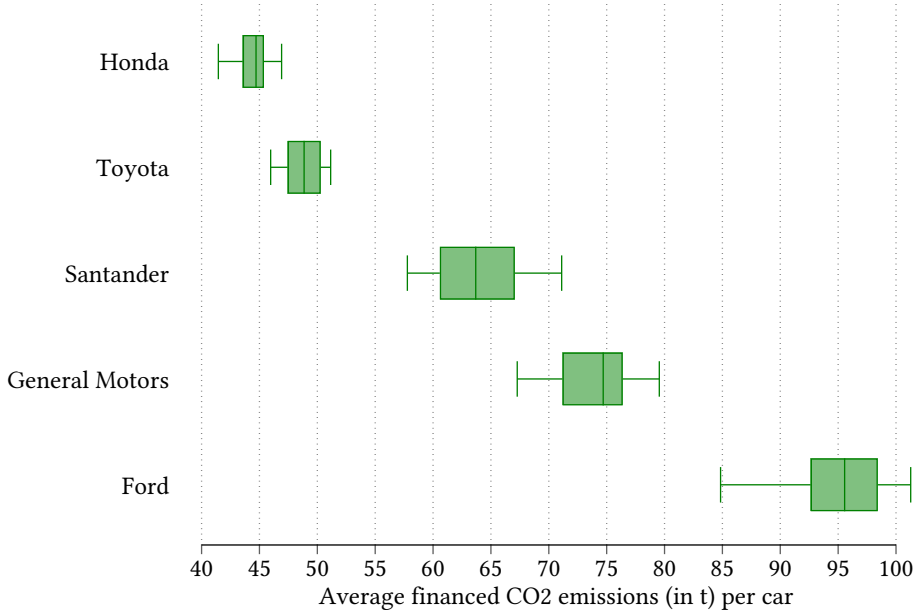


Figure 1: Dispersion of CO_2 Emissions across all ABS Pools for select issuers

an investor has the choice to invest in an auto ABS which finances high-emissions vehicles (e.g., Ford) or an auto ABS which finances low-emissions vehicles (e.g., Toyota).

The differences in emissions across deals are largely explained by the vehicle type composition of the collateral pool. For example, across 21 ABS deals issued by Honda, the average share of passenger vehicles in the collateral pools is 40.2%, the average share of SUVs is 58.3%, and the average share of trucks is 1.5%. In contrast, across the 16 ABS deals issued by Ford, these shares are 11.8%, 38.7%, and 49.5%, respectively. The vehicle type composition for captive lenders is related to the products offered by their manufacturing parents and consumer demand for them. The high share of trucks and the high-emissions of Ford's auto ABS reflect the fact that Americans have bought more F-Series trucks than any other vehicle for more than four decades[10].

Environmentally conscious investors can use the fact that differences in emissions across ABS deals are explained by the vehicle type composition to distinguish brown from green auto ABS. Table 1 reports coefficients from a regression of average tCO_2 emissions per car on the share of trucks in the collateral pool. For each 1 percentage point increase in the truck share,

[10] Kelley Blue Book: Ford F-150 Retakes Best-Selling Vehicle Crown

Table 1: Regression of average tCO_2 Emissions per car on vehicle types

	Constant	Truck share	SUV share	Adj. R^2	N	Avg. tCO_2/car
β	44.108***	1.019***	0.210***	0.746	281	70.514
(se) or sd	(1.879)	(0.036)	(0.040)			15.550

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

the average amount of CO_2 per car increases by 1.021 ton. The vehicle type shares are reported in the securities prospectus and are available as a variable in the loan-level data and enable investors to distinguish brown from green auto ABS.

3 Data

This section describes the loan-level data used to construct the measures of greenness and provides information about the issuance level data I use in the empirical tests.

3.1 Loan-level Data

The loan-level data is from the Securities and Exchange Commission (SEC) form ABS-EE. Form ABS-EE is part of the post-financial crisis reporting requirements under Regulation AB, which went into effect on November 23, 2016. Under the reporting requirement, all prospectuses for public offerings of asset-backed securities must submit loan-level information in electronic format. Further, every month loan-level information about the performance of the loan pool are to be provided.

The loan-level data covers all public auto ABS issued from 2017 to 2022; consisting of approximately 18.9 million unique loans from 281 ABS deals issued by 22 issuers[11]. The data contains information on the originator, borrower, and collateral of each loan. Appendix Table A1 presents summary statistics of loan-level data. The average borrower in my sample finances $26,000, at 90% loan-to-value, at an 7.6% interest rate for 67 months. Their credit

[11]These issuers are: Ally Financial, AmeriCredit, BMW Financial, Capital One Bank, CarMax, Carvana, Exeter Finance, Fifth Third Bank, Ford Credit, GM Financial, Honda Finance, Hyundai Capital, JM Family (WOART), JM Family (WOSAT), Mechanics Bank, Mercedes-Benz Financial Services, Nissan Finance, Santander Bank (DRIVE), Santander Bank (SDART), Toyota Motor Credit, USAA Federal Savings Bank, and Volkswagen Credit.

score is 712 and their monthly payment to income ratio is 0.08. The vehicle the average borrower is financing is worth $27,733.

3.2 Emissions Data

The data on CO_2 emissions comes from the EPA. This data is matched on the make, model, and model year to the loan-level data. Estimates of survival-weighted vehicle miles traveled (SVM) by vehicle type are from the EPA CAFE standard emissions simulator. The SVM estimates vary by vehicle type and assume up to 40 years of useful life. The EPA estimates that trucks and SUVs have higher SVM than sedans and compact cars. Appendix Figure B2 plots SVM curves by vehicle type.

The average vehicle in my sample is driven for 202,963 miles, of which 162,450 miles are financed by the proceeds of the auto ABS. Over its total lifetime, the vehicle emits 79t of CO_2, of which 63t are financed. There is considerable heterogeneity among emissions of the collateral since the sample includes fully-electric vehicles, compact cars, SUVs, as well as pick-up trucks and other high-emissions vehicles. The 10 most popular vehicles in my sample show this heterogeneity. These are, in order, Toyota Camry (sedan, on average 60t of CO_2 emissions over full lifetime), Toyota RAV4 (SUV, 73t), Toyota Corolla (sedan, 53t), Nissan Rogue (SUV, 62t), Chevrolet Silverado (truck, 120t), Honda Civic (sedan, 51t), Nissan Altima (sedan, 59t), Honda CR-V (SUV, 65t), Honda Accord (sedan, 62t), and Ford F-150 (truck, 114t)[12].

3.3 ABS Deal Data

I collect information about the structure of each deal from prospectuses filed with the SEC (Form 424H, 424B5, and FWP). These contain information about the characteristics of each deal and its tranches, e.g., issue date, ratings, coupon, spreads, issuance amounts, weighted average life (WAL), and book running banks. I calculate issuance spreads as the difference between the issuance yields and yield curve estimates of Filipović, Pelger, and Ye (2022) by matching the maturity against the WAL of a tranche. Table 2 shows summary statistics for the A-2 tranches of each deal. The average deal size is $1.2bn, of which 30% are associated with the A-2 tranche. The average spread is 42bps with a WAL of one year. Around 42% of

[12]The top 10 models make up 28% of observations and track the lists of most sold cars in the US over the past 5 years.

Table 2: Summary Statistics Issuance-Level Data (A-2 Tranche)

	Mean	SD	Median	Min	Max	N
Total Deal Size ($ m)	1,234.93	344.47	1,250.00	367.31	2,663.82	281
Tranche Size ($ m)	366.71	131.99	362.00	42.40	746.94	281
Weight. Avg. Life (years)	0.98	0.32	1.01	0.37	3.50	281
Spread (bps)	41.68	29.10	32.29	6.13	194.22	281
Coupon (%)	1.91	1.30	1.86	0.14	5.81	281
Subprime ABS	0.28	0.45	0.00	0.00	1.00	281
Captive Lender	0.44	0.50	0.00	0.00	1.00	281
Number of Loans	66,953	25,500	66,011	15,212	180,352	281
Avg. Loan-to-Value	0.92	0.04	0.92	0.80	0.98	281
Avg. Credit Score	706.20	74.85	738.43	564.98	788.46	281
Avg. Interest Rate (%)	7.64	5.86	4.46	1.38	21.35	281
Avg. % of Balance Outstanding	0.90	0.07	0.91	0.74	1.00	281
Avg. Warehousing Time (months)	9.54	4.38	9.19	1.33	21.07	281
Expected tCO2 per $100,000	292.83	51.42	296.31	161.51	456.16	281
Expected tCO2 per car	70.51	15.55	67.61	42.94	125.73	281
Financed tCO2 per $100,000	219.58	40.08	211.15	107.10	311.78	281
Financed tCO2 per car	58.01	12.76	54.49	40.54	101.27	281

deals are issued by captive lenders and around 28% are sub-prime. The average deal finances around 67,000 cars. A $100,000 investment finances 220t of CO_2 over the remaining life of the collateral. The average car an investor finances emits approximately 58t of CO_2 .

4 Identification Strategy

My strategy for causal identification of non-pecuniary preferences is supported by two points. Firstly, high levels of standardization and the short-term, safe-asset nature of the securities make it unlikely that unobserved heterogeneity at the issuer or security level contaminate my estimates. The seniority structure of auto ABS deals ensures that prepayment is the main risk factor for senior tranches. Secondly, I provide evidence that the CO_2 emissions of the collateral do not predict prepayment, suggesting that observed effects are driven by non-pecuniary preferences rather than differences in prepayment risk which correlate with CO_2 emissions. The differences in the cost of capital I recover therefore likely reflect the market's taste for greener deals.

While standard no-arbitrage asset pricing theory dictates that identical cash flows from

brown and green securities should be priced identically in equilibrium (Ross, 1976), there is a growing literature that argues otherwise. Both theoretical (e.g. Pástor et al., 2021) and empirical contributions (e.g. Zerbib, 2019) show effects of non-pecuniary preferences for green assets on asset prices. Studying the effect of non-pecuniary preferences on asset prices empirically is challenging since unobserved heterogeneity can create spurious correlation between greenness and cash flows. For example, environmental regulations, like a carbon tax, can correlate with both greenness and expected cash flows. This complexity is further compounded by the lack of objective and standardized measures of greenness (Berg, Koelbel, and Rigobon, 2022), making it challenging to quantify the environmental impact of an investment and differentiate between green and non-green securities.

I address these challenges by studying the effects of greenness on the cost of capital in the auto ABS market. The high levels of standardization in the security design mitigate the usual challenges that unobserved heterogeneity at the security level poses. Moreover, auto ABS are issued by special purpose vehicles which are set up to be bankruptcy remote from their sponsoring entity. Even if the sponsor (e.g, General Motors) enters bankruptcy, holders of auto ABS and the collateral-pools they finance are protected. This makes it unlikely that unobserved heterogeneity at the sponsor level influences expected cash flows across auto ABS[13].

The main risk factor for senior tranches of auto ABS is prepayment. By allocating cash flows and credit risks of the underlying assets in a specific manner across tranches, the design of these deals limits other sources of risk, such as credit default or collateral performance, primarily to subordinate tranches. Time tranching involves dividing cash flows from underlying assets into different tranches based on payment priority, ensuring senior tranches receive cash flows before subordinate tranches. While usual auto loans have a five-year duration, time tranching ensures that the senior tranches I study have a weighted average life of only one year. Risk tranching involves assigning different levels of credit risk to tranches, with senior tranches having lower default risk and subordinate tranches facing higher default risk.[14] My

[13]Regular stocks and bonds are susceptible to unobserved heterogeneity, as firm-specific characteristics like debt covenants (e.g. Green, 2019), regulatory environments (e.g. Seltzer et al., 2022), employee satisfaction (e.g. Edmans, 2011), or management quality (e.g. Bennedsen, Nielsen, Pérez-González, and Wolfenzon, 2007) influence firm-performance and security prices.

[14]If a borrower in the collateral pool defaults on their loan, the car gets repossessed and sold again. To holders of senior tranches the default, repossession, and subsequent recovery constitutes "involuntary" prepayment. The difference in outstanding balance and recovery value is born by the most junior tranche. Historic recovery values are around 60% and 45% for prime and subprime loans, respectively (Structured Finance Association).

analysis focuses on senior tranches which are AAA-rated by at least two rating agencies. The threshold at which pool-level credit losses start to impact these tranches is around 50% because of risk tranching[15].

The granularity of the loan-level data allows me to control for predictors and ex-post realizations of prepayment risk. The determinants of prepayment risk include borrower characteristics (e.g., creditworthiness, income level), loan characteristics (e.g., loan-to-value ratio, loan term), and prevailing market conditions (e.g., interest rates, unemployment rate). Importantly, prepayment risk is primarily determined by borrower and loan characteristics rather than being influenced by the underlying collateral's greenness. Carbon dioxide emissions provide a quantifiable and reliable indicator of the environmental impact associated with an investment in auto ABS. They do not, however, predict prepayment at the intensive or extensive margin as I show below. This implies that any observed impact on asset prices and financing costs can be attributed to a preference for the greenness of the collateral, rather than greenness being a proxy for prepayment risk.

I use the unique features of the auto ABS market to test if non-pecuniary preferences lower the cost of capital for green securities using the following specification:

$$\log\left(\text{Issuance Spread}\right)_{bt} = \alpha\,\text{Green}_b + \zeta' X_b + \gamma_t + \varepsilon_{bt} \tag{2}$$

for bond tranche b issued in year-month t. Green_b is an indicator variable equal to one if the collateral pool's CO_2 emissions per USD are below the 50th percentile of all auto ABS deals and zero otherwise[16]. The coefficient of interest is α, which reflects the premium investors are willing to accept for a green security. The specification in eq. (2) is consistent with the literature on the green premium which uses a discrete definition of greenness (Zerbib, 2019, Larcker and Watts, 2020). I additionally test the following specification which uses a continuous definition of greenness:

$$\log\left(\text{Issuance Spread}\right)_{bt} = \beta \log\left(\text{CO}_2\text{ Emissions}\right)_b + \zeta' X_b + \gamma_t + \varepsilon_{bt} \tag{3}$$

[15] Assuming a zero percent recovery value.

[16] Appendix Table A4 shows a covariate balance table. Other than the sorting variable, only the average credit score and interest rate of green and brown deals have statistically different means at the 5% level. The differences in means, however, are less than one standard deviation.

The coefficient of interest is β, which reflects the elasticity of issuance spreads with respect to CO_2 emissions of the underlying collateral pool. I use both the financed emissions of a collateral pool as tCO_2 per USD principal and tCO_2 per car. In robustness checks I also use the average miles-per-gallon, average truck share, average GHG rating, as well as the level of CO_2 emissions of the collateral pools.

The identifying assumption in both equation (2) and (3) is that CO_2 emissions are uncorrelated with the error term conditional on traditional risk factors: once prepayment risk is accounted for, assignment of CO_2 emissions is "as good as random." This identification assumption allows me to answer the following questions: once an investor can be sure that their loan will be paid back at the agreed time, do the CO_2 emissions of the collateral influence their choice to invest – and is this choice reflected in price? A natural null hypothesis is that the greenness of the collateral does not influences prices, i.e. $\alpha = \beta = 0$. That is, $\alpha < 0$ (or $\beta > 0$) is evidence that investors prefer greener collateral. Conversely, $\alpha > 0$ (or $\beta < 0$) provides evidence that investors prefer collateral which is less environmentally-friendly.

The identification assumption would be violated if vehicles with higher CO_2 emissions carry higher prepayment risk after accounting for borrower or loan characteristics. If CO_2 emissions were correlated with better performance, it would provide a risk-based rationale for investors to assign a lower cost of capital to browner securities. Risk compensation, however, is the central tenant of classical asset pricing theories and different from the "warm-glow" that an investor might receive from holding greener assets (e.g., as in Pástor et al., 2021).

I am able to directly test the identifying assumption at the collateral-pool and loan-level since auto ABS file monthly performance reports. I derive ex-post performance variables from these reports that account for future realizations of prepayment risk. Specifically, I examine two measures of voluntary and involuntary prepayment at the pool-level: the average realized difference in monthly absolute prepayment speed (APS) compared to the prospectus assumption, and the average realized percentage of loans that are more than 30 days delinquent.

The CO_2 emissions of the collateral do not predict prepayment at the collateral pool level or the individual loan-level. Appendix Table A2 shows estimates from univariate regressions of financed tCO_2 per USD and per car on measures of voluntary and involuntary prepayment. The estimates show that CO_2 emissions are not a significant predictor of pool-level performance even unconditional on traditional risk factors. The estimates are noisy, close to zero, switch signs across tCO_2 per USD and tCO_2 per car and the R^2 is less than 1% in most specifi-

cations. Appendix Table A3 provides a similar analysis at the individual-loan-level. At the individual level the analysis tests whether CO_2 emissions have predictive power for prepayment at both the extensive and intensive margins, taking into account predictors of prepayment risk at the borrower and loan-level. The estimated coefficients of CO_2 emissions are again close to zero. Moreover, the standardized CO_2 coefficients are an order of magnitude smaller than usual predictors of prepayment risk like credit score or interest rate. I take the results of these two exercises as evidence that the identifying assumption holds and proceed as if assignment of CO_2 emissions are "as good as random" conditional on traditional risk factors.

The vector X_b contains controls for the current market conditions, tranche characteristics, and collateral pool characteristics. To control for current market conditions, X_b includes estimates of the high-quality corporate yield curve for maturities of 6 and 12 months, the level of the VIX, the standard deviation of the VIX during the month before issuance, and the level of the 5-year breakeven inflation rate at issuance. The included tranche characteristics are the weighted average life, default attachment point, and the issuance size. The vector X_b also contains collateral pool controls that are predictors and ex-post realizations of prepayment risk. These are the collateral pool averages of LTV, credit scores, remaining loan balance share, interest rate, warehousing time, realized difference in APS, and the share of loans which are more than 30 days delinquent after 12 months. Collateral pool variables derived from individual loans are weighted by outstanding loan amount. Collateral pool and tranche characteristic have individual slopes across subprime and prime issues to account for potentially different pricing relevance. All control variables are in logs to achieve a more symmetric and normally distributed data distribution. The specifications further control for subprime issue fixed effects, as well as fixed effects relating to the pricing speeds of the different deals, the assumed prepayment speed (APS) fixed effects. I interact the APS fixed effects with the weighted average life of the tranche to account for potential "ramp-up" periods in which prepayments increase and then level off to their APS. Standard errors are clustered at the year-month level to account for common variation among spreads caused by market conditions.

5 Results: Impact of Greenness on the Cost of Capital

In this section I document the main results that (i) if one does not carefully account for the risk-factors which drive prices in the ABS market, greener securities seem to have a statisti-

cally significant lower cost of capital, (ii) that this relationship flips around once I account for security design and prepayment risk. I document that the finding that senior tranches of auto ABS with higher CO_2 emissions have a lower cost of capital is robust to alternative specifications, samples, and measures of greenness. I also evidence that this relationship is not driven by a taste for higher CO_2 emissions *per se* but by preferences for issuers which produce high emissions collateral. Auto ABS of these issuers tend to have higher ESG and credit ratings, which also induces a lower cost of capital. The fact that ESG ratings are positively correlated with CO_2 emissions at the collateral pool attenuates the influence of CO_2 on issuance spreads. Lastly, this section discusses the results and relates them to literature on the green premium in debt markets.

5.1 The Influence of CO2 Emissions on the Cost of Capital

Panel A, Panel B, and Panel C of Table 3 present estimates of the pricing models of equation (2) and equation (3), respectively. The *naive* specification in column (1) only controls for current market conditions as well as year-month FE. In Panel A, the estimated difference in the cost of capital of auto ABSs with emission-intensity below the median is 25%[17] and statistically significant ($t = 6.3$). This translate to a green premium of 10bps. This estimate is almost twice as large as other findings in the literature. Baker et al. (2022) estimate a green premium of about 6bps in a large sample of municipal and corporate green bonds. Similarly, columns (1) in Panel B shows an estimated elasticity of issuance spreads with respect to CO_2 emissions of 1.000. This implies that Ford, whose auto ABS have on average 25% higher emissions per USD than Honda, has to pay 25% higher issuance spreads.

However, the specification of columns (1) fail to account for observed and unobserved heterogeneity which influences the speed by which the collateralized loans prepay. The following columns address this incrementally by absorbing fixed effects and adding controls for predictors and ex-post realizations of prepayment risk. Columns (2) adds a fixed effect for the assumed prepayment speed (APS) to control for heterogeneity in the design of the securities. The estimated coefficients in Panel A and B are approximately 2/3 smaller but still imply a significant lower cost of capital of around 10%. Columns (3) further adds subprime fixed effects as well as other tranche characteristics like attachment point, weighted average life, and

[17] $25\% \approx 100(\exp\left(-0.286 - 0.0455^2/2\right) - 1)$, see Halvorsen and Palmquist (1980).

Table 3: Estimates of the influence of CO_2 emissions on the Cost of Capital

Panel A: Semi-Elasticity with respect to Green indicator

	(1)	(2)	(3)	(4)	(5)
		log(Issuance Spread)			
Green (tCO2/USD<p50)	-0.286***	-0.0936**	0.00245	0.0465	0.0646*
	(0.0455)	(0.0303)	(0.0278)	(0.0328)	(0.0278)
Year-Month FE, market conditions	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	No	Yes	Yes	Yes	Yes
Other tranche characteristics	No	No	Yes	Yes	Yes
Ex-ante prepayment controls	No	No	No	Yes	Yes
Ex-post prepayment controls	No	No	No	No	Yes
Adj. R^2	0.821	0.912	0.952	0.965	0.970
Within R^2	0.154	0.065	0.475	0.615	0.671
Observations	276	276	276	276	276

Panel B: Elasticity with respect to financed tCO_2 per USD

	(1)	(2)	(3)	(4)	(5)
		log(Issuance Spread)			
log(Financed tCO2 per USD)	1.000***	0.274**	-0.0179	-0.154	-0.190+
	(0.0999)	(0.0845)	(0.0936)	(0.117)	(0.112)
Year-Month FE, market conditions	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	No	Yes	Yes	Yes	Yes
Other tranche characteristics	No	No	Yes	Yes	Yes
Ex-ante prepayment controls	No	No	No	Yes	Yes
Ex-post prepayment controls	No	No	No	No	Yes
Adj. R^2	0.844	0.912	0.952	0.965	0.970
Within R^2	0.264	0.063	0.475	0.616	0.670
Observations	276	276	276	276	276

Panel C: Elasticity with respect to financed tCO_2 per car

	(1)	(2)	(3)	(4)	(5)
		log(Issuance Spread)			
log(Financed tCO2 per car)	0.0532	-0.145+	-0.265***	-0.202**	-0.197**
	(0.107)	(0.0785)	(0.0667)	(0.0701)	(0.0676)
Year-Month FE, market conditions	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	No	Yes	Yes	Yes	Yes
Other tranche characteristics	No	No	Yes	Yes	Yes
Ex-ante prepayment controls	No	No	No	Yes	Yes
Ex-post prepayment controls	No	No	No	No	Yes
Adj. R^2	0.790	0.910	0.957	0.966	0.971
Within R^2	0.008	0.050	0.523	0.628	0.677
Observations	276	276	276	276	276

Standard errors in parentheses clustered at year-month (K=71). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001.

the issuance size. The three levels of fixed effects, year-month, subprime, and APS explain more than 90% of variation in the issuance spreads. In Panel A and B, the estimated green premium shrinks towards zero, flips sign, and becomes statistically indistinguishable from zero. Columns (4) further adds predictors of prepayment speed like the average credit score, warehousing time, loan-to-value, interest rate, and outstanding balance. Columns (5) introduces two ex-post performance variables. These are the difference of the realized to the assumed prepayment speed and the realized share of loans that are more than 30 days delinquent. Adding these variables increases the within R^2 by around 10%.

After I account for differences in security design and prepayment speeds the elasticities of issuance spreads with respect to CO_2 emissions in Table 3 are negative and statistically significant, implying that higher emissions deals have a lower cost of capital. Column (5) in Panel A shows that auto ABS with CO_2 emissions below the median have a 6.5% *higher* cost of capital. The elasticity estimate of -0.19 in column (5) Panel B implies that moving from the 20th percentile (189t) of CO_2 emissions intensity to the 80th percentile (262t), a 1.85sd distance equivalent to comparing Toyota to Ford auto ABS, leads to a reduction in the cost of capital of 7% ($\approx -0.19 \times (262/189 - 1)$). The estimated elasticity of issuance spread with respect to CO_2 emissions per car in column (5) in Panel C similarly shows that issuance spreads are 0.2% lower per 1% higher emissions.

5.2 The Influence of ESG ratings on the Cost of Capital

The identification strategy outlined above also allows me to test if the ESG and credit ratings of the issuers of auto ABS influence issuance spreads. The identifying assumption in both equation (2) and (3) is that ESG and credit ratings of issuers are uncorrelated with the error term conditional on traditional risk factors. This assumption should hold by construction since both ESG and credit ratings are irrelevant for the payoffs of securitized loans. However, they might reflect "tastes" for certain issuers over others[18].

I use the ESG and credit rating of Standard & Poor's as well as Refinitiv's ESG rating of auto ABS issuers to test for the presence of taste for certain issuers. Both ESG and credit

[18]These tastes may be caused by frictions or portfolio constraints of some investor groups. For example, California's Government Code section 53601(o) mandates that government agencies only invest in auto ABS if the issuer has at least a credit rating of A, even though auto ABS are bankruptcy remote from their issuer. In my sample only 15 out of 22 issuers have a S&P credit rating and only 9 have an A rating at some point during the sample period from 2016 to 2022.

Table 4: Elasticity of issuance spreads with respect to ESG rating and credit rating of auto ABS issuers

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
				log(Issuance Spread)					
S&P ESG Rating Issuer	-0.177***	-0.162**	-0.164**						
	(0.0506)	(0.0522)	(0.0505)						
Refinitiv ESG Rating Issuer				-0.365***	-0.339**	-0.359***			
				(0.104)	(0.122)	(0.101)			
S&P Credit Rating Issuer							-0.107+	-0.145*	-0.121*
							(0.0537)	(0.0563)	(0.0567)
Financed tCO2 per car		-0.0825			-0.0499			-0.192*	
		(0.0664)			(0.0827)			(0.0825)	
Expected tCO2 per USD			-0.105			-0.164			-0.140
			(0.102)			(0.117)			(0.113)
Year-Month FE, market conditions	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.976	0.976	0.976	0.976	0.976	0.977	0.978	0.979	0.979
Within R^2	0.749	0.752	0.752	0.752	0.753	0.758	0.744	0.755	0.748
Observations	235	235	235	235	235	235	190	190	190

Standard errors in parentheses clustered at year-month (min K=58). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table 5: Correlation between environmental impact of auto ABS and ESG ratings of Issuers

	Refinitiv ESG Rating	S&P ESG Rating	CO_2 per car	CO_2 per USD	Avg. MPG	Avg. Truck %
Refinitiv ESG Rating	1.00					
S&P ESG Rating	0.73	1.00				
CO_2 per car	0.41	0.31	1.00			
CO_2 per USD	0.11	0.20	0.58	1.00		
Avg. MPG	0.23	0.08	0.80	0.44	1.00	
Avg. Truck %	0.16	0.08	0.77	0.42	0.75	1.00

Notes: Correlation among N=235 observations for which ESG ratings of issuers are available. MPG is multiplied by (-1) such that higher values correspond to worse environmental performance.

ratings are unrelated to the risk factors and environmental impact of collateralized auto loans. Note that less than half of my sample are issued by captive auto lenders of manufacturers. Table 2 shows that the majority of auto ABS in my sample is issued by independent banks, auto finance companies, or used vehicle retailers. It would be surprising if the ESG rating of a large multi-national bank like Santander, which originated 43 out of 281 deals in my sample, would incorporate the environmental impact of auto ABS collateral pools.

Columns (1) and columns (4) of Table 4 show that both higher ESG rating by S&P and Refinitiv cause lower issuance spreads. The estimates show elasticities between 0.18 and 0.37. Column (7) shows that credit ratings of the issuer cause lower issuance spreads with an elasticity of 0.11. Columns (2), (3), (5), and (6) include ESG ratings of the issuers as well as CO_2 emissions of the collateral pools. The estimated elasticities of the ESG ratings remain statistically significant and stable across specification. The elasticities of CO_2 emissions, however, are attenuated by up to 50% and estimated imprecisely.

Table 5 provides the explanation for this finding. It shows the correlations between the environmental impact of auto ABS and the ESG ratings of issuers. While the ESG rating of S&P and Refinitiv are positively correlated with each other, they are also positively correlated with the CO_2 emissions of collateral pools. It suggests that the reason that browner auto ABS deals have a lower cost of capital is because they are correlated with high ESG ratings and investors seem to pay a premium for securities with high ESG rating. Similarly, investors seem to reward issuers of auto ABS with higher credit ratings with lower spreads. Both these findings confirm the presence of tastes for certain issuers over others.

5.2.1 Robustness

The result that environmentally worse auto ABS have a lower cost of capital is robust to using alternative measures, tranches, and specifications.

Lower cost of capital for brown auto ABS is robust alternative measures of Greenness I perform a series of robustness test using different measures of greenness for each auto ABS deal. The other measures of greenness are (i) the expected tCO_2 per USD, (ii) the expected tCO_2 per car, (iii) the average MPG of the cars in the collateral pool, (iv) the average truck share in the collateral pool, and (v) an independently constructed measure of greenness by the Kroll Bond Rating Agency (KBRA)[19]. The measures (i) to (v) are not perfectly correlated ($0.1 < |\rho| < 0.67$) with the measure of financed tCO_2 from Table 3 and therefore provide an independent signal about the relative greenness of each auto ABS deal. While measures (ii) to (iv) are highly correlated with the KBRA measure ($|\rho| > 0.75$), it serves as a good robustness check since it is independently constructed, publicly available, and a salient feature of the US car market.

Appendix Table A5 shows that the results do not change qualitatively when using different measures of relative greenness. All specifications show that browner auto ABS have a lower cost of capital. Quantitatively, most estimates imply an elasticity of around -0.2, close to the estimates of the main results in Table 3.

Lower cost of capital of brown auto ABS holds across the capital structure The analysis above uses A-2 tranches as those all have similar characteristics across different deals; low credit risk, clean-up call options are non-binding, and have the highest observation count. However, the results are robust to the choice of other AAA-rated tranches. Appendix Table A6 reports regression results across all AAA-rated tranches. The table shows qualitatively and quantitatively similar results to the main results in Table 3 except for money-market tranches. The estimates elasticities of issuances spreads with respect to emissions are also near -0.2 in other tranches. This implies that the lower cost of capital for environmentally worse auto ABS is scaling through the entire capital structure of these deals.

[19]KBRA released free reports in March 2021 (KBRA, 2021) and July 2022 (KBRA, 2022) mapping the EPA's vehicle GHG scores to 247 auto ABS deals. The EPA's GHG scores, ranging from 1 to 10 with higher values indicating lower emissions, are displayed on window labels attached to each new car in the US since 2013.

Lower cost of capital for brown auto ABS is unrelated to credit quality A potential concern is that I use both prime and sub-prime auto ABS. Differences in CO_2 emissions might be correlated with unobserved borrower characteristics that are related to loan-quality. For example, subprime borrowers more often buy used cars and likely find it harder to refinance their loans or trade in their vehicle than prime borrowers. Appendix Table A7 shows that the result is robust to using prime auto ABS deals only. The estimated elasticities of issuances spreads with respect to emissions are between 0.16 and 0.19 in prime auto ABS, similar to the main result in Table 3. This alleviates potential concerns that the estimated elasticities are contaminated by unobserved heterogeneity causing the greenness of the collateral to be related to borrower quality.

5.3 Discussion

The result that higher CO_2 emissions are associated with a lower cost of capital, stands in contrast to prior results in the literature. For example, Pástor et al. (2022) document a near-arbitrage opportunity in which German government green "twin" bonds trade on average at 5bps lower yields relative to their non-green twins. There are several possible explanations for the difference between my results and prior results in the literature. I discuss the three most pertinent: (i) differences in issuance and market size, (ii) high information acquisition cost, and (iii) "green window dressing".

Firstly, differences in issuance and market size could explain the differences in pricing of green assets. ESG capital is still scarce and the chance of being the marginal investor is decreasing in issuance size. Several studies looking at pricing differences in the green municipal-bond market have found a green premium (e.g., Baker et al., 2022). Painter (2020) reports that the average issuance size for municipal bonds over $1m is $13.4m. The average issuance size in the A-2 tranches of auto ABS I analyze is an order of magnitude larger at $367m. If investors with environmental preferences are infra-marginal in setting prices, we would still expect a higher share of green assets on their balance sheet. I explore this channel by examining the balance sheets of ESG mutual funds in Section 6.

Secondly, high information acquisition could imply unawareness among ESG investor and thus explain the absence of a green premium. There are, however, free reports by rating agencies (e.g, KBRA, 2022, 2021) and ESG industry groups that discuss the greenness of auto ABS.

Moreover, as I document in Table 1, the emissions content of an ABS pool is related to the share of trucks it finances. This variable is featured in security prospectuses and accessible in the loan-level data. The availability of free information and the presence of a sufficient statistic make it unlikely that information acquisition play a large role in explaining the differences in findings.

Lastly, the ostensibly conflicting results can also be reconciled by a green window dressing effect in which securities that are especially marketed as green trade at a premium while securities that markedly differ in their true greenness but are not marketed as such do not. My estimates in Table 4 show that high ESG ratings of auto ABS issuers are associated with a lower cost of capital for auto ABS. Moreover, high ESG ratings tend to coincide with high CO_2 emissions of collateral pools. The positive correlation between the ESG ratings and CO_2 attenuates my results; supporting the "green window dressing" explanation. I leave the exploration of this channel for future research.

6 Mutual Funds Holdings of auto ABS

In seminal work, Hong and Kacperczyk (2009) argue that norm-constrained investors, like ESG mutual funds, are more likely to invest in responsible securities compared to unconstrained investors. I therefore turn to portfolio holdings of mutual funds to analyze if the CO_2 emissions of the collateral influence ESG mutual funds' decision to invest.

I document two counterintuitive facts: ESG mutual funds (i) hold positions across the full distribution of CO_2 emissions and (ii) ESG funds invest more in higher-emission deals relative to non-ESG funds. The latter relationship is confounded by the fact that ESG ratings of auto ABS issuers are positively correlated with the CO_2 emissions of collateral pools. Once this confounding factor is taken into account, it emerges that ESG funds invest more in brown auto ABS whose issuers have high ESG ratings. This highlights the misalignment between the environmental criteria emphasized by ESG ratings of issuers and the actual greenness of the collateral pools.

Table 6: Summary Statistics of Holdings Data

	Mean	SD	Median	Min	Max	N
Portfolio Share	0.17	0.22	0.09	0.00	1.40	11,325
Coupon Yield	1.90	1.21	1.92	0.00	6.51	11,325
Tranche Size ($m)	263.32	168.75	230.50	8.51	746.94	11,325
Weighted Average Life	2.35	0.99	2.38	0.11	5.06	11,325
Subprime ABS	0.41	0.49	0.00	0.00	1.00	11,325
Financed tCO2 per USD	227.80	37.88	226.88	118.20	311.78	11,325
Financed tCO2 per car	59.65	12.18	58.31	40.54	101.27	11,325

6.1 Data

I obtain mutual funds holdings from the SEC Form N-PORT over the period from 2019-Q3 to 2022-Q4. In the data I keep the first observation for which a mutual fund reports a position in a senior tranche of an auto ABS. I trim portfolio shares at the 1 % to account for outliers and reporting errors[20]. Table 6 provides summary statistics of the mutual fund holdings data. I observe 266 individual auto ABS deals in the holdings data[21]. Mutual funds are among the largest group of investors in auto ABS. The mutual funds hold an average of 23% of issuance per deal on their balance sheets. This share can go as high as 85% for some senior tranches. Other notable institutional investors in senior auto ABS are insurance companies, corporate treasuries, and local governments.

In the holdings data, I identify ESG mutual funds in two ways: (i) by their name using key words[22] like "sustainable", "ESG", or "climate" and (ii) using a list of "Sustainable Investment Mutual Funds and ETFs" offered by institutional member firms of "The Forum for Sustainable and Responsible Investing" (USSIF)[23]. I identify 35 ESG funds (and 787 non-ESG funds) which hold at least one position in an auto ABS tranche over the sample period (32 ESG funds hold at least two positions). These ESG funds make up 25% of the 143 ESG-bond funds which I am able to identify in the N-PORT filings.

[20] The results are qualitatively similar to using 2%, 5%, or not trimming.

[21] The discrepancy to the 281 auto ABS deals in Section 5 is explained by the shorter sample period for which N-PORT filings are available.

[22] The key word list contains the following words: "green", "climate", "esg", "sustainable", "enviroment", "responsible", "impact", "catholic", "social", "sri", "csr", "community", and "justice".

[23] https://charts.ussif.org/mfpc/

6.2 ESG Mutual Funds' approach to auto ABS

The prospectuses of ESG mutual funds often detail their investment selection approach with regards to asset-backed securities. For example,

" [...] When evaluating securitized debt securities [...] , the Adviser generally considers the issuer's ESG rating along with ESG factors related to the underlying pool of assets, such as energy efficiency and environmental impact of the underlying assets[...]"

– ESG Mutual Fund Prospectus

The fact that ESG mutual funds discuss the "environmental impact of the underlying assets" shows awareness of the different environmental impacts that collateral pools can have. Interestingly, they also mention the ESG ratings of issuer of auto ABS. I therefore also test which dimension, ESG rating of issuers or actual CO_2 emissions, influences an ESG funds choice to invest.

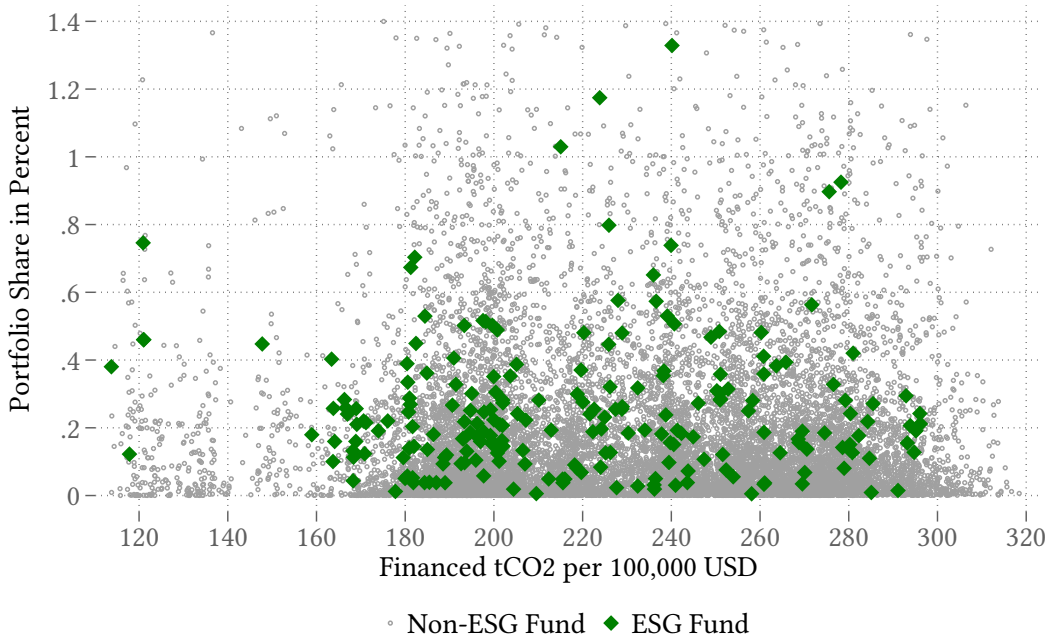
6.3 Identification Strategy

To test if ESG funds tilt their portfolio towards greener auto ABS, I estimate a reduced form asset demand function in the spirit of Koijen and Yogo (2019). I use the following reduced form model for portfolio shares in tranche t of auto ABS deal b by mutual funds i in year-quarter r:

$$\log\left(\text{Portfolio Share}\right)_{itrb} = \alpha\left(\text{ESG Fund}_i \times \text{Green}_b\right) + \gamma_i + \gamma_b + \gamma_r + \zeta' X_t + \varepsilon_{itrb} \tag{4}$$

where Green_b is an indicator variable equal to one if the collateral pool's CO_2 emissions per car are below the 50th percentile of all auto ABS deals; γ_i are fund fixed effects; γ_b are auto ABS deal fixed effects; and γ_r reporting year-quarter fixed effects. The vector X_t contains tranche characteristics which have different slopes for subprime and prime issues. These characteristics are the weighted average life, size, and annualized yield of the tranche. The coefficient of interest is α which captures the effect of variation in greenness of the collateral on portfolio shares by ESG-funds relative to non-ESG funds.

I leverage the fact that mutual funds hold positions in multiple auto ABS to tightly identify if ESG funds invest more capital in environmentally friendly auto ABS compared to non-ESG funds. By using a stringent set of fixed effects I estimate ESG-funds preferences from variation in greenness of multiple auto ABS deals held by ESG funds relative to non-ESG funds during the same reporting period. The empirical specifications include collateral pool and fund fixed effects and thus absorb characteristics and preferences of each mutual fund and the specific features of each auto ABS deal. By including these fixed effects, the model effectively controls for as much of the unobserved heterogeneity across



Note: X-axis is jittered with normally distributed noise for readability.

Figure 2: Portfolio Shares of Mutual Funds in Auto ABS

collateral pools and funds as possible. The relative difference in preferences for green assets by ESG versus non-ESG funds, however, remains identified. Standard errors are clustered at the fund-level to account for common variation at the fund-level.

6.4 Results

Figure 2 plots portfolio shares in auto ABS of mutual funds against financed CO_2 emissions per \$100,000. The graph shows that ESG mutual funds hold positions across the full distribution of CO_2 emissions. That ESG funds have positions across the full distribution is surprising since common ESG strategies imply either outright exclusions of brown assets or best-in-class investment. Counterintuitive to best-in-class investment, Figure 2 also shows that ESG funds hold similar or higher shares in browner auto ABS.

Table 7 provides estimates of the relationship between greenness and ESG ownership using the specification of equation (4). The estimated coefficients in Panel A of Table 7 indicate that the greenest 50% auto ABS receive between 15% to 26% less capital from ESG funds relative to non-ESG funds. Column (1) of Panel A controls for tranche characteristics and absorbs fund and deal fixed effects. The α

Table 7: Reduced Form Demand of Mutual Fund for Auto ABS – Environmental Impact

Panel A: Green Indicator

	(1)	(2)	(3)	(4)
		log(Portfolio Share)		
Green (tCO2<p50)=1 × ESG Fund=1	-0.162*	-0.210*	-0.221*	-0.280
	(0.0760)	(0.0915)	(0.0911)	(0.213)
Tranche controls	Yes	Yes	Yes	Yes
Fund FE, Deal FE	Yes	Yes	Yes	Yes
Year-Quarter FE	No	Yes	Yes	Yes
Tranche FE	No	No	Yes	No
Cusip FE	No	No	No	Yes
Adj. R^2	0.829	0.834	0.835	0.867
Observations	11207	11207	11207	11102

Standard errors clustered at fund-level (min K=714).

Panel B: other measures of relative *greenness*

	(1)	(2)	(3)	(4)
		log(Portfolio Share)		
ESG Fund=1 × Financed tCO2 per car	0.162*			
	(0.0687)			
ESG Fund=1 × Avg. MPG		0.206***		
		(0.0605)		
ESG Fund=1 × Truck Share			0.136	
			(0.138)	
ESG Fund=1 × Avg. GHG Rating (KBRA)				0.233**
				(0.0777)
Tranche controls	Yes	Yes	Yes	Yes
Fund FE, Deal FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Tranche FE	Yes	Yes	Yes	Yes
Adj. R^2	0.835	0.835	0.835	0.832
Observations	11202	11202	11202	10432

Coefficients are standardized to unit variances in Panel B. Average MPG and GHG Rating are negated, higher values correspond to worse environmental performance. Standard errors clustered at the fund-level (min K=685). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001

coefficient is identified from variation in greenness of auto ABS deals held by ESG funds relative to non-ESG funds. Column (2) adds reporting-quarter fixed effects such that α is restricted to variation within the same reporting period. Column (3) adds additional tranche FE effects to control for differential demand across tranches (e.g., A-2 vs. A-3 tranche). Finally, Column (4) uses CUSIP fixed effects to control for unobserved heterogeneity at the security level.

Panel B of Table 7 presents similar estimates using other measures of greenness. All estimate of α in Panel B are identified from variation in greenness of auto ABS deals held by ESG funds relative to non-ESG funds during the same reporting period. The measure of greenness in column (1) is the average financed tCO_2 per car; average miles per gallon in column (2); share of trucks in the collateral pool in column (3); and the average GHG rating provided by KBRA in column (4). In all columns the estimated coefficients of α are positive and of similar magnitude. The estimates in column (1) of Panel B imply that moving from the 10th percentile of average financed tCO_2 per car to the 90th percentile (a 2.75 standard deviation distance equivalent to comparing Honda to Ford auto ABS) results in 0.45sd higher portfolio shares for a ESG funds compared to a non-ESG funds.

The correlation between ESG ratings and CO_2 emissions is even stronger in the mutual fund holdings data as Appendix Table A8 shows. Table 8 therefore repeats the reduced form demand estimation of Table 7 but controlling for the ESG ratings of auto ABS issuers. The estimates in columns (1) and (4) show that both the ESG rating of S&P and Refinitiv are highly correlated with the differential demand by ESG funds relative to non-ESG funds. Moreover, columns (2), (3), (5), and (6) show that the differential demand for auto ABS by issuers with high ESG-ratings attenuates the estimates of differential demand for high-emissions auto ABS by up to 50%.

ESG funds are holding more auto ABS of issuers with high ESG-rated relative to non-ESG funds. That by itself is not surprising. However, the ESG ratings of issuers are positively correlated with the CO_2 emissions of collateral. ESG funds therefore inadvertently hold higher portfolio shares in environmentally less friendly auto ABS than non-ESG funds.

6.5 Discussion

This subsection discusses the results that ESG mutual funds (i) hold positions across the full distribution of CO_2 emissions and (ii) that ESG funds invest more in higher-emission deals, and (iii) that this relationship is attenuated when one takes into account that CO_2 emissions of collateral pools and ESG ratings of issuers are positively correlated.

Money managers report that the leading ESG criteria are climate change and CO_2 emissions[24].

[24] USSIF (2022): Sustainable Investing – Money Managers 2022

Table 8: Reduced Form Demand of Mutual Funds for Auto ABS – ESG Ratings

	(1)	(2)	(3)	(4)	(5)	(6)
	log(Portfolio Share)					
ESG Fund=1 × S&P ESG Rating	0.127**	0.0854+	0.121**			
	(0.0448)	(0.0480)	(0.0454)			
ESG Fund=1 × Refinitiv ESG Rating				0.171***	0.137**	0.167***
				(0.0486)	(0.0508)	(0.0474)
ESG Fund=1 × Financed tCO2 per car		0.113			0.0794	
		(0.0787)			(0.0744)	
ESG Fund=1 × Financed tCO2 per USD			0.0448			0.0426
			(0.0683)			(0.0620)
Fund FE	Yes	Yes	Yes	Yes	Yes	Yes
ABS Deal FE	Yes	Yes	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Tranche FE	Yes	Yes	Yes	Yes	Yes	Yes
Tranche controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.836	0.836	0.836	0.836	0.836	0.836
Observations	9987	9987	9987	9987	9987	9987

Standard errors in parentheses clustered at fund-level (min K=639). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001
Coefficients are standardized to unit variances.

However, my findings suggest that ESG mutual funds are not investing in the most environmentally-friendly securities but securities whose issuers have higher ESG ratings, even if those securities have a higher CO_2 emissions intensity. This finding raises questions about the effectiveness of ESG investment strategies in addressing environmental externalities. Fund managers may need to re-evaluate their investment processes and methods to ensure that they are fulfilling their obligations to promote environmentally sustainable investing. Fund managers may also need to look for alternative investments that are better aligned with their ESG objectives and consider the environmental impact of their investments more closely in order to meet the expectations of their investors.

The findings further suggest that there may be a need for more clarity and transparency in ESG labeling and investment processes. Policymakers may need to provide greater guidance to the financial sector on what constitutes environmentally sustainable investing, and to ensure that ESG labels accurately reflect the environmental impact of investments.

ESG regulation in the United States is still in its infancy. The SEC has taken some steps to provide guidance to the financial industry to ensure that ESG labels accurately reflect the environmental impact of investments. The SEC has issued several statements and guidance documents related to ESG investing and has encouraged companies to provide more comprehensive and transparent disclosure

of their ESG practices and impact. In Europe, there are similar efforts to ensure the accuracy of ESG labeling. The European Union has adopted the Sustainable Finance Disclosure Regulation (SFDR), which requires companies and market participants to provide more comprehensive and transparent disclosure of their sustainability risks, impacts, and objectives. Emiris, Harris, and Koulischer (2023) examine the impact of the SFDR on portfolio allocation and ESG fund flows. Their findings indicate that the regulation led to increased flows to ESG funds, particularly among investors with stronger environmental preferences, and that funds with higher initial uncertainty about their sustainability benefited the most from the disclosure.

7 Conclusion

This paper sheds new light on the question if greener securities have a lower cost of capital in debt markets. I compare the cost of capital for green and brown automobile asset-backed securities and provide a causal estimate of the premium investors are willing to pay for greener securities. The results show that auto ABS with CO_2 emissions above the median have a 6.5% lower cost of capital. The result that brown auto ABS have a lower cost of capital is robust to alternative measures of greenness, samples, and specifications.

The paper further provides evidence that ESG mutual funds hold positions across the full distribution of CO_2 emissions, indicating that they are not investing in the most environmentally-friendly securities. The paper shows that ESG funds tend to hold higher portfolio shares in high-emissions auto ABS compared to non-ESG funds. This is surprising since greener auto ABS are not riskier and neither do browner securities pay higher yields. However, issuers of browner securities tend to have better ESG ratings which seem to drive the differential demand of ESG mutual funds.

These findings have implications for retail investors of ESG mutual funds, fund managers, and policy makers. The results suggest that commonly used ESG ratings are uninformative about the environmental impact of certain securities and that ESG mutual funds are not investing in the most environmentally-friendly securities. These findings raise questions about the effectiveness of ESG investment strategies in addressing environmental externalities.

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Table A1: Summary Statistics Loan-Level Data

	Mean	SD	Median	Min	Max	N
Original Interest Rate	7.62	6.95	4.99	0.00	30.00	18,863,466
Original Loan Amount ($)	26,154.45	12,488.05	23,925.29	518.03	248,681.95	18,863,467
Original Loan Term (months)	67.56	8.68	72.00	7.00	96.00	18,863,467
Credit Score	711.48	101.21	722.00	250.00	900.00	18,029,902
Payment-to-Income Ratio	0.08	0.05	0.08	0.00	0.79	18,644,373
Income Verified	0.09	0.29	0.00	0.00	1.00	18,863,467
Loan-to-Value	0.90	0.16	1.00	0.01	1.00	18,862,126
Outstanding Balance Ratio	0.83	0.24	0.93	0.00	1.00	18,863,463
Vehicle Value Amount ($)	27,723.22	13,358.86	25,251.00	0.00	1,084,455.00	18,863,464
Vehicle Age (Years)	2.69	2.53	2.00	0.00	35.00	18,863,467
Used Vehicle	0.46	0.50	0.00	0.00	1.00	18,863,467
SVM, Financed	162,526.16	39,846.07	173,251.82	254.15	240,728.61	18,863,467
SVM, Total	202,941.11	17,235.53	207,738.97	189,173.82	240,728.61	18,863,467
tCO2, total Lifetime	78.87	33.08	72.43	0.00	538.75	18,863,467
tCO2, remaining Lifetime	62.97	31.90	56.71	0.00	538.75	18,863,467
tCO2, financed remaining Lifetime	47.05	29.53	44.64	0.00	538.75	18,862,122

A Appendix

Table A2: CO_2 Emissions and Ex-post Performance of Collateral Pools

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Δ Realized APS				Realized % loans delinquent 30d+			
Financed tCO2 per USD	0.0762	0.0888			-0.0372	0.0436		
	(0.0635)	(0.0603)			(0.0540)	(0.0370)		
Financed tCO2 per car			-0.0170	-0.0283			0.0806	0.00490
			(0.0528)	(0.0529)			(0.0538)	(0.0377)
Subprime FE	No	Yes	No	Yes	No	Yes	No	Yes
Adj. R^2	0.00581	0.0209	0.000289	0.0139	0.00139	0.625	0.00650	0.623
Observations	281	281	281	281	281	281	281	281

Robust standard errors in parenthesis. + p<0.10, * p<0.05, ** p<0.01, *** p<0.001. Coefficients are standardized to unit variances.

Table A3: Test of Identifying Assumption in Loan-Level Data

	𝟙(Prepayed, 12m)		% prepayed, 12m		log (USD Prepayed)	
	(1)	(2)	(3)	(4)	(5)	(6)
Financed tCO2 per USD	0.00325*		0.00174+		0.00194+	
	(0.00135)		(0.000962)		(0.000991)	
Subprime=1 × Financed tCO2 per USD	0.00811**		0.00956***		0.00935***	
	(0.00259)		(0.00203)		(0.00206)	
Financed tCO2		0.00380*		0.00203+		0.00227+
		(0.00158)		(0.00113)		(0.00116)
Subprime=1 × Financed tCO2		0.00946**		0.0112***		0.0109***
		(0.00303)		(0.00237)		(0.00241)
Loan-to-value	-0.0475***	-0.0453***	-0.0564***	-0.0552***	-0.0564***	-0.0551***
	(0.00415)	(0.00400)	(0.00277)	(0.00262)	(0.00277)	(0.00263)
Subprime=1 × Loan-to-value	0.0113+	0.0169**	0.0509***	0.0575***	0.0509***	0.0573***
	(0.00659)	(0.00601)	(0.00354)	(0.00341)	(0.00354)	(0.00341)
Credit score	0.0172***	0.0172***	-0.0112***	-0.0112***	-0.0113***	-0.0113***
	(0.00491)	(0.00491)	(0.00217)	(0.00217)	(0.00218)	(0.00218)
Subprime=1 × Credit score	0.0760***	0.0760***	0.0340***	0.0340***	0.0340***	0.0340***
	(0.00625)	(0.00625)	(0.00359)	(0.00359)	(0.00360)	(0.00360)
Interest rate	0.0996***	0.0996***	0.0841***	0.0841***	0.0844***	0.0844***
	(0.00696)	(0.00696)	(0.00301)	(0.00301)	(0.00302)	(0.00302)
Subprime=1 × Interest rate	0.0305***	0.0305***	-0.0150***	-0.0150***	-0.0149***	-0.0149***
	(0.00893)	(0.00893)	(0.00443)	(0.00443)	(0.00444)	(0.00444)
Warehousing time	-0.0605***	-0.0605***	-0.0996***	-0.0996***	-0.0999***	-0.0999***
	(0.00572)	(0.00572)	(0.00438)	(0.00438)	(0.00438)	(0.00438)
Subprime=1 × Warehousing time	0.0468***	0.0468***	-0.0106+	-0.0106+	-0.0107+	-0.0107+
	(0.00616)	(0.00616)	(0.00577)	(0.00577)	(0.00578)	(0.00578)
Year-State FE	Yes	Yes	Yes	Yes	Yes	Yes
Subprime FE and slopes	Yes	Yes	Yes	Yes	Yes	Yes
Used vehicle FE	Yes	Yes	Yes	Yes	Yes	Yes
Originator FE	Yes	Yes	Yes	Yes	Yes	Yes
Subvention, exemption FE	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.0373	0.0373	0.191	0.191	0.573	0.573
Within R^2	0.008	0.008	0.070	0.070	0.477	0.477
Observations	15,981,964	15,981,964	3,423,059	3,423,059	3,423,059	3,423,059

Standard errors in parentheses clustered at year-state and ABS pool-level (min K=282). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001
Coefficients are standardized to unit variances.

Table A4: Covariate Balance Test of *Green* and *Brown* auto ABS Deals

Variable	(1) Brown (CO2>=p50) Mean/(SE)	(2) Green (CO2<p50) Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Financed tCO2 per USD	251.861	187.059	64.802***
	(2.187)	(1.761)	
Yield Curve 6m	0.013	0.014	-0.001
	(0.001)	(0.001)	
Yield Curve 12m	0.014	0.016	-0.002
	(0.001)	(0.001)	
VIX	17.377	21.661	-4.284
	(0.577)	(0.593)	
5 Year Breakeven Inflation	1.863	2.202	-0.339
	(0.038)	(0.055)	
Attachment Point	0.504	0.463	0.041*
	(0.006)	(0.006)	
Weight. Avg. Life	0.930	1.030	-0.100
	(0.026)	(0.028)	
Tranche Size	336.219	397.411	-61.192
	(10.636)	(11.062)	
Loan-to-Value	0.927	0.909	0.017
	(0.003)	(0.003)	
Mean Credit Score	673.859	738.773	-64.913***
	(6.297)	(5.018)	
Mean Interest Rate	9.862	5.406	4.456*
	(0.506)	(0.403)	
Mean % of outstanding	0.923	0.873	0.049
	(0.005)	(0.006)	
Warehousing Time (Months)	8.185	10.909	-2.723
	(0.334)	(0.369)	
Number of observations	141	140	281

Note: Similar to the regression analysis the t-test includes clean-up call FE, APS FE, captive and subprime FE, and year-month FE. * p<0.05, ** p<0.01, *** p<0.001

Table A5: Elasticity of issuance spreads with respect to different measures of *Greenness*

	(1)	(2)	(3)	(4)	(5)
			log(Issuance Spread)		
log(Expected tCO2 per USD)	-0.229*				
	(0.104)				
log(Expected tCO2 per car)		-0.199**			
		(0.0673)			
log(Avg. MPG)			-0.0100+		
			(0.00534)		
log(Avg. Truck Share)				-0.191*	
				(0.0929)	
log(Avg. GHG Rating) [KBRA]					-0.0429+
					(0.0256)
Year-Month FE, Market conditions	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.970	0.971	0.970	0.970	0.968
Within R^2	0.675	0.678	0.670	0.671	0.694
Observations	276	276	276	276	243

Standard errors in parentheses clustered at year-month (min K=71). + p<0.10, * p<0.05, ** p<0.01, *** p<0.001
MPG and GHG Rating are multiplied by (-1), higher values correspond to worse environmental performance.

Table A6: Estimates of elasticity of issuance spreads with respect to emissions in other senior tranches

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread	(5) Issuance Spread	(6) Issuance Spread
	A-1 Tranche		A-3 Tranche		A-4 Tranche	
Financed tCO2 per USD	0.115		-0.233**		-0.266***	
	(0.196)		(0.0836)		(0.0719)	
Financed tCO2 per car		-0.0228		-0.157*		-0.138*
		(0.160)		(0.0615)		(0.0576)
Year-Month FE	Yes	Yes	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.950	0.950	0.964	0.964	0.981	0.980
Within R^2	0.415	0.413	0.604	0.603	0.465	0.445
Observations	234	234	272	272	190	190

Standard errors in parentheses clustered at year-month.

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A7: Estimates of elasticity of issuance spreads with respect to emissions in prime auto ABS

	(1) Issuance Spread	(2) Issuance Spread	(3) Issuance Spread	(4) Issuance Spread
Financed tCO2 per USD	-0.158			
	(0.108)			
Expected tCO2 per USD		-0.190^{+}		
		(0.104)		
Financed tCO2 per car			-0.160*	
			(0.0626)	
Expected tCO2 per car				-0.163*
				(0.0620)
Year-Month FE	Yes	Yes	Yes	Yes
Market condition controls	Yes	Yes	Yes	Yes
Assumed prepayment speed FE	Yes	Yes	Yes	Yes
Other tranche characteristics	Yes	Yes	Yes	Yes
Ex-ante prepayment controls	Yes	Yes	Yes	Yes
Ex-post prepayment controls	Yes	Yes	Yes	Yes
Adj. R^2	0.973	0.973	0.974	0.974
Within R^2	0.637	0.641	0.645	0.645
Observations	190	190	190	190

Standard errors in parentheses clustered at year-month (min K=70).

+ p<0.10, * p<0.05, ** p<0.01, *** p<0.001

Table A8: Correlation of ESG ratings and measures of environmental impact in N-PORT holdings data

	Refinitiv ESG Rating	S&P ESG Rating	Financed tCO2/car	Financed tCO2/USD	Avg. MPG	Truck %	GHG Rating
Refinitiv ESG Rating	1.00						
S&P ESG Rating	0.85	1.00					
Fin. tCO2/car	0.50	0.41	1.00				
Fin. tCO2/USD	0.36	0.34	0.54	1.00			
Avg. MPG	0.32	0.25	0.83	0.42	1.00		
Truck %	0.33	0.24	0.83	0.33	0.86	1.00	
GHG Rating	0.27	0.15	0.75	0.19	0.86	0.85	1.00

MPG and GHG Rating are multiplied by (-1) such that higher values are environmentally worse. $N = 9447$.

B Online Appendix (for online publication only)

(a) Santander SDRIVE 2021-4 Subprime Issue

Pricing $1.8bn Santander Drive Auto Receivables Trust 2021-4

Issuer: Santander Consumer USA
Lead Managers: Citi(str.), JPM, and SIS
DE&I Co-managers: AmeriVet Securities, Great Pacific Securities, Mischler Financial Group

Anticipated Capital Structure:

CL	OFF. AMT	WAL	F/M	L.FNL	BENCH	SPRD	YLD%	CPN	PX
A-1	$222.40	0.13	F1+/P-1	11/15/2022	Intl.	7	0.16802	0.16802	100.00000
A-2	$543.10	0.64	AAA/Aaa	08/15/2024	EDSF	20	0.380	0.37	99.99378
A-3	$292.37	1.43	AAA/Aaa	08/15/2025	EDSF	15	0.517	0.51	99.99081
B	$288.61	2.10	AA/Aaa	06/15/2026	IntS	30	0.887	0.88	99.98887
C	$243.48	2.80	A/Aa2	02/16/2027	IntS	47	1.271	1.26	99.97902
D	$241.38	3.58	BBB/Baa2	10/15/2027	IntS	70	1.685	1.67	99.98569
E	$131.18	3.97	NR/B2	<<NOT OFFERED>>					

(b) CarMax 2019-1 Prime Issue

$1.5bln CarMax (CARMX) 2019-1
JOINT BOOKRUNNERS : Credit Suisse (str), Barclays, Wells Fargo
CO-MANAGERS : MUFG, Scotia, SMBC, TD

CLS	$AMT(MM)	WAL	S&P/FITCH	P.WIN	L.FNL	BNCH	SPRD	YLD%
A1	277.000	0.28	A-1+/F1+	1-7	01/2020	IntL -	1	2.78007
A2A	412.000	1.16	AAA/AAA	7-22	07/2022	EDSF +	31	3.045
A2B	100.000	1.16	AAA/AAA	7-22	07/2022	1mL +	31	
A3	493.900	2.64	AAA/AAA	22-43	03/2024	IntS +	40	3.074
A4	107.910	3.84	AAA/AAA	43-48	08/2024	IntS +	65	3.283
B	42.170	3.98	AA/AA	48-48	11/2024	IntS +	85	3.479
C	39.910	3.98	A/A	48-48	01/2025	IntS +	115	3.779
D	27.110	3.98	BBB/BBB	48-48	08/2025	IntS +	145	4.079

* Exp. Settle: 01/23/19
* First Pay Date: 02/15/19
* Px Speed: 1.30% ABS to 10% Call
* Timing: PRICED
* Format: Public/SEC
* ERISA: Yes
* Min Denoms: $5k by $1k
* B&D: Credit Suisse

Figure B1: Examples of Typical auto ABS Deals

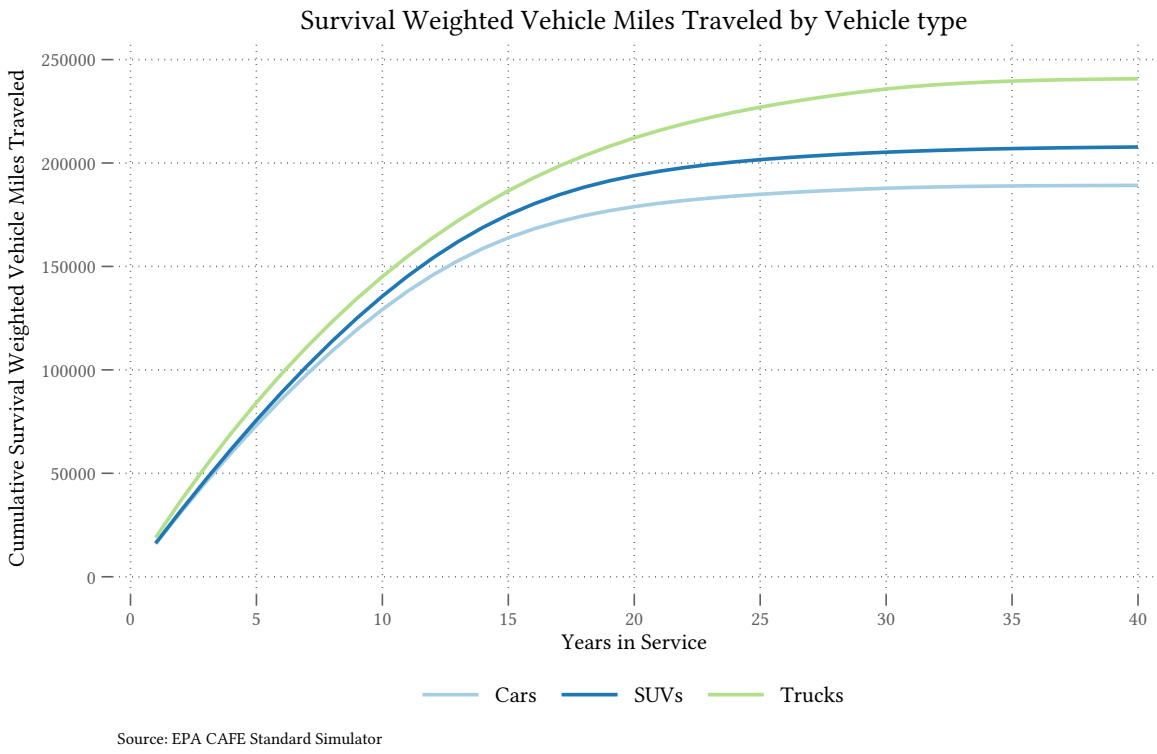


Figure B2: Survival-Weighted Vehicle Miles by Vehicle Type